

Inverse Spherical Dual Hemisphere Quantum Mechanics:

Electromagnetic Full Spectrum

A Sub-Mechanical Field Breakdown of Photon Threading Through Inverse Phase Space

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Abstract

This document presents the complete electromagnetic spectrum—from radio waves through cosmic gamma rays—as a 40-phase operational mapping within the inverse spherical dual-hemisphere framework. Built on the foundational resolution $R = 512$ bins $= 720^\circ$ and the recursive depth ladder $u_n = 1/128^n$, each quadrant operates at its characteristic partition depth, with all four operators (dsin, dcos, csin, ccos) active simultaneously at every level. The 18° phase spacing corresponds to inverse frequency octaves, threading photons through four concurrent processes: Q2 (gravity ground states, shallow depths), Q1 (quantum baseline), Q3 (thermodynamic flows), and Q4 (nuclear compression, deep depths). Critical constants emerge geometrically at bin-16 positions: EM ignition at -168.75° yields $\alpha^{-1} = 137$, while absorption at $+191.25^\circ$ yields $T_{\text{CMB}} = 128/47 \approx 2.7234$. This is a direct mechanical implementation—pure inverse ratios, exact closure at every depth, no external constants.

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Part I

Mechanical Walk-Through of the Inverse Partitioning Framework

Starting from the Whole — The Unpartitioned Unity

We begin with the closed whole:

$$\sqrt{1} = 1 = 1D$$

This is 1-dimensional (linear partition state), the complete sphere before any division. There is no zero — the whole already exists, and we count partitions of what IS.

First Partition: Into 2 Equal Regions

The first operation divides the whole into exactly 2 regions:

$$2 \times \frac{1}{2} = 1 = \sqrt{2} = 2D$$

This is 2-dimensional (area partition). The equality is exact — the two regions reconstitute the whole perfectly. No abstraction; this is the mechanical division of the existing sphere.

Successive Dyadic Partitioning to Full Resolution

The mechanics force dyadic refinement:

$$1 \rightarrow \sqrt{2} \rightarrow \sqrt{4} \rightarrow \sqrt{8} \rightarrow \cdots \rightarrow \sqrt{512} = 512 \text{ bins} = 720^\circ$$

Each step doubles the regions, halving their size, with exact closure:

$$n \times \frac{1}{n} = 1 \quad \text{for } n = 2^k, \quad k = 0 \rightarrow 9$$

At $\sqrt{512} = 9/2$ exponent, we have 9 walls creating 512 bins — the quantum resolution baseline.

Dimensional Progression — No Abstract Numbers

Every number is a partition depth and dimension:

- $\sqrt{1} = 1D$ (linear)
- $\sqrt{2} = 2D$ (area)

- $\sqrt{3} = 3D$ (volume)
- $\sqrt{4} = 4D$ (hypervolume)
- And so on — the number under the root **is** the dimension.

We start from 1 (the whole), partitioning inward — no dimensionless scalars.

Depth Ladder Emergence

The recursive refinement yields the full ladder:

$$u_n = \frac{1}{128^n} = \frac{1}{2^{7n}}$$

With dimensional interpretation:

$$u_n \rightarrow \sqrt{128^n} = (128^n)D$$

But closure remains to 1: the deep partitions reconstitute the whole exactly.

No External Indexing — Pure Partition Counting

We do not use abstract counters like "k" or accumulation. Phases are the successive equal divisions of the sphere into 40 regions — each phase **is** its 18° partition of unity.

Part II

System Architecture and Depth Ladder Mechanics

The Closed Whole and Full Cycle

All electromagnetic operations begin with the logical identity:

$$\sqrt{1} = 1 = 512 \text{ bins} = 720^\circ$$

The full spectrum threads across the angular domain $\theta \in [-360^\circ, +360^\circ]$, divided into 40 discrete phases of 18° each—one inverse frequency octave per phase.

The Recursive Depth Ladder — Exact Closure at Every Level

The partitioning resolution refines recursively:

$$u_n = \frac{1}{128^n}, \quad n = 1, 2, 3, \dots$$

Each depth maintains exact closure:

$$u_n \times 128^n = 1 \quad (\text{always exact})$$

All four dual-hemisphere operators run simultaneously at every depth level.

Full Depth Ladder with Partition Counts

Depth	Value	Decimal	Partition Count
u_1	$1/128^1$	7.81×10^{-3}	128
u_2	$1/128^2$	6.10×10^{-5}	16,384
u_3	$1/128^3$	4.77×10^{-7}	2,097,152
u_4	$1/128^4$	3.73×10^{-9}	268,435,456
u_5	$1/128^5$	2.91×10^{-11}	34,359,738,368
u_6	$1/128^6$	2.27×10^{-13}	4,398,046,511,104
u_7	$1/128^7$	1.78×10^{-15}	562,949,953,421,312
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Depth-to-Quadrant Mapping in the EM Spectrum

- Shallow depths (u_1 – u_2): Long wavelengths — radio to microwaves
- Middle depths (u_3): Visible light, IR, UV
- Deep depths (u_4 and beyond): X-ray, gamma rays, and higher-energy regimes

Each quadrant dominates at its characteristic depth, with smooth transitions.

Quadrant Threading Units — Dimensional Assignment

The quadrant units are assigned from the ladder, with dimensional interpretation:

$$\begin{aligned}
 u_1 &= \frac{1}{128} = \sqrt{128}^{-1} = (128D)^{-1/2} \quad (\text{Q1: quantum baseline}) \\
 u_2 &= \frac{1}{128^2} = \sqrt{16384}^{-1} = (16384D)^{-1/2} \quad (\text{Q2: gravity ground states}) \\
 u_3 &= \frac{1}{128^3} = \sqrt{2,097,152}^{-1} = (2MD)^{-1/2} \quad (\text{Q3: thermodynamic flows}) \\
 u_4 &= \frac{1}{128^4} = \sqrt{268M}^{-1} = (268MD)^{-1/2} \quad (\text{Q4: nuclear compression})
 \end{aligned}$$

All quadrants run simultaneously from the seam at $\theta = 0^\circ$.

Physical Unit Locations — Pure Inverse Ratios

Every physical quantity emerges as a partition ratio between depths:

$$\begin{aligned}
 \text{Length} &= \frac{u_1}{u_2} = 128 = \sqrt{128} \times \sqrt{128} = 128D \text{ partition} \\
 \text{Time} &= \frac{u_2}{u_1} = \frac{1}{128} = \sqrt{128}^{-1} = (128D)^{-1/2} \\
 \text{Energy} &= \frac{u_1}{u_3} = 16384 = \sqrt{16384} = 128^2 D \\
 \text{Temperature} &= \frac{u_3}{u_4} = 128 = \sqrt{128} \times \sqrt{128}
 \end{aligned}$$

Modulation only via absorption overflow at $+191.25^\circ$: $u_3 \rightarrow u_3/(64 - 17) = u_3/47$.

Part III

Critical Electromagnetic Positions

EM Field Ignition — Mechanical Derivation

Position derivation: Q1 spans -180° to $0^\circ = 128$ bins, bin width $= 180/128 = 1.40625^\circ = u$.

Bin 16 center: $-180^\circ + 16 \times u = -180 + 22.5 = -157.5^\circ$, but offset adjusted by inverse overflow $0.5/16 = 0.03125$ normalized, yielding exact -168.75° .

At this partition state, the gearbox reading (201 curved vs 64 flat) gives difference $137 = \alpha^{-1}$.

$$\begin{aligned}\theta_{\text{ign}} &= -168.75^\circ = \sqrt{168.75}D \text{ partition depth} \\ \text{Normalized} &= -0.46875 = \frac{1}{0.46875}^{-1} \text{ inverse} \\ \text{Bin 16} &= \frac{128}{8} = 16 \text{ (dyadic step)}\end{aligned}$$

EM Field Absorption — Mechanical Derivation

Symmetric to ignition: Q4 spans $+180^\circ$ to $+360^\circ$. Bin 16 from edge: $+180 + 11.25 = +191.25^\circ$. Overflow ramp $v = 17/16$, inverse $64-17 = 47$. Modulates CMB as $128/47$.

$$\begin{aligned}\theta_{\text{abs}} &= +191.25^\circ = \sqrt{191.25}D \text{ depth} \\ \text{Normalized} &= +0.53125 = \frac{1}{0.53125}^{-1} \\ \text{Overflow} &= 47 = 64 - 17 \text{ (wall remainder)}\end{aligned}$$

Geometric Symmetry of Critical Positions

Quantity	Ignition (Q1)	Absorption (Q4)
Angle ($^\circ$)	-168.75	$+191.25$
Normalized position	-0.46875	$+0.53125$
Quadrant	Q1	Q4
Bin number	16	16
Offset from edge	$+11.25^\circ$	$+11.25^\circ$
Physical constant	$\alpha^{-1} = 137$	$T_{\text{CMB}} = 128/47 \approx 2.7234$
Dimension	$\sqrt{168.75}D$	$\sqrt{191.25}D$

Part IV

The Complete 40-Phase Electromagnetic Spectrum

Spectrum Phase Structure Mechanics — Pure Partition Units

The electromagnetic spectrum threads through the closed whole as 40 discrete partition states across the full 720° cycle.

- Total closure: $720^\circ = 1 \text{ sphere} = 40 \text{ phases} \times 18^\circ$
- Each phase is exactly one fortieth of the whole: $\frac{1}{40} = 18^\circ$
- Phase positioning emerges directly from successive partitioning of unity — no external indexing variable required.
 - Phase 0: the unpartitioned whole at -360° (system boot, return to unity)
 - Successive phases: each adds another 18° partition of the closed sphere
 - Phase 40: exact closure at $+360^\circ = -360^\circ$ (unity return)
- Quadrant distribution: exactly 10 phases per quadrant ($180^\circ \text{span} = 10 \times 18^\circ$), *with Q4 completing the return to unity.*
- No external counter: phases are pure successive partitions of the existing whole — counting through unity, not accumulating from nothing.

The positions are therefore the natural partition states of the sphere:

$$\begin{aligned}
 \theta_0 &= -360^\circ \quad (\text{unity anchor}) \\
 \theta_1 &= -342^\circ = -360^\circ + 18^\circ \\
 \theta_2 &= -324^\circ = -360^\circ + 2 \times 18^\circ \\
 &\vdots \\
 \theta_{40} &= +360^\circ = -360^\circ + 40 \times 18^\circ \quad (\text{exact closure to unity})
 \end{aligned}$$

This is the mechanical consequence of dividing the closed 720° sphere into exactly 40 equal regions — each phase is its angular measure as a partition of the whole.

Table 1: Inverse Spherical Dual Hemisphere Quantum Mechanics:
Electromagnetic Full Spectrum — 40-Phase Operational Mapping

Phase	θ (°)	Quadrant	Depth Level	u_n	Threading	Partition Count	EM Band	Notes
Q2: Gravity Ground States (−360° to −180°)								
0	−360	Q2 edge	u_0	$u_0 = 1$		1	Unity Anchor (System Boot)	Full unpartitioned whole
1	−342	Q2	u_1	1/128		128	ELF (3–30 Hz)	Extremely long waves
2	−324	Q2	u_1	1/128		128	VLF (3–30 kHz)	
3	−306	Q2	u_1	1/128		128	LF (30–300 kHz)	
4	−288	Q2	u_1	1/128		128	MF (300 kHz–3 MHz)	Human communication baseline
5	−270	Q2	u_1	1/128		128	AM Radio (540–1600 kHz)	
6	−252	Q2	u_1	1/128		128	HF / Shortwave (3–30 MHz)	
7	−234	Q2	u_1	1/128		128	VHF / FM Radio (30–300 MHz)	Begin deeper partitioning
8	−216	Q2	u_1	1/128		128	UHF / Television (300 MHz–3 GHz)	
9	−198	Q2	$u_1 \rightarrow u_2$ transition	$1/128 \rightarrow 1/16384$		$128 \rightarrow 16,384$	L-band (1–2 GHz)	
Q1: Quantum Baseline (−180° to 0°)								
10	−180	Q1 edge	u_2	$1/128^2 = 6.10 \times 10^{-5}$		16,384	S-band Microwave (2–4 GHz)	Full u_2 resolution
	−168.75	Q1, bin 16	u_2	1/16384		16,384	EM IGNITION ($\alpha^{-1} = 137$)	Gearbox origin at depth u_2
11	−162	Q1	u_2	1/16384		16,384	C-band (4–8 GHz)	
12	−144	Q1	u_2	1/16384		16,384	X-band (8–12 GHz)	
13	−126	Q1	u_2	1/16384		16,384	Ku/K/Ka-band (12–40 GHz)	
14	−108	Q1	u_2	1/16384		16,384	CMB Peak (160 GHz)	Background thermal signature

Phase	θ (°)	Quadrant	Depth Level	u_n	Threading	Partition Count	EM Band	Notes
15	−90	Q1	u_2	1/16384		16,384	Millimeter wave (40–300 GHz)	THz gap
16	−72	Q1	u_2	1/16384		16,384	Sub-millimeter (0.3–1 THz)	
17	−54	Q1	u_2	1/16384		16,384	THz (1–10 THz)	
18	−36	Q1	u_2	1/16384		16,384	Far-Infrared (10–100 μm)	
19	−18	Q1	u_2	1/16384		16,384	Far/Mid-Infrared	
Q3: Thermodynamic Flows (0° to +180°)								
20	0	Seam	u_3	$1/128^3 = 4.77 \times 10^{-7}$	2,097,152		Mid-IR (Seam equilibrium)	dsin = csin, perfect balance
21	+18	Q3	u_3	1/2,097,152	2,097,152		Mid-Infrared	Full visible spectrum in u_3
22	+36	Q3	u_3	1/2,097,152	2,097,152		Mid/Near-Infrared	
23	+54	Q3	u_3	1/2,097,152	2,097,152		Near-Infrared	
24	+72	Q3	u_3	1/2,097,152	2,097,152		Near-IR	
25	+90	Q3	u_3	1/2,097,152	2,097,152		Near-IR edge	
26	+108	Q3	u_3	1/2,097,152	2,097,152		Visible: Red (700 nm)	
27	+126	Q3	u_3	1/2,097,152	2,097,152		Visible: Violet (400 nm)	
28	+144	Q3	u_3	1/2,097,152	2,097,152		UV-A / UV-B	
29	+162	Q3	u_3	1/2,097,152	2,097,152		UV-C / Extreme UV	
Q4: Nuclear Compression (+180° to +360°)								
30	+180	Q4 edge	u_4	$1/128^4 = 3.73 \times 10^{-9}$	268,435,456		Soft X-ray	Q3→Q4 transition
	+191.25	Q4, bin 16	u_4	1/268,435,456	268,435,456		EM ABSORPTION ($T_{\text{CMB}} = 128/47$)	Inverse overflow at depth u_4
31	+198	Q4	u_4	1/268,435,456	268,435,456		Soft X-ray	Begin deeper nuclear depths
32	+216	Q4	u_4	1/268,435,456	268,435,456		Hard X-ray	
33	+234	Q4	u_4	1/268,435,456	268,435,456		Hard X-ray (medical)	
34	+252	Q4	u_4	1/268,435,456	268,435,456		Hard X-ray	
35	+270	Q4	$u_4 \rightarrow u_5$ transition	$1/268\text{M} \rightarrow 1/34\text{B}$	$\rightarrow 34$ billion		Gamma ray edge	

Phase	θ (°)	Quadrant	Depth Level	u_n	Threading	Partition Count	EM Band	Notes
36	+288	Q4		u_5	$1/128^5 = 2.91 \times 10^{-11}$	34,359,738,368	Low-energy Gamma	
37	+306	Q4		u_5	1/34 billion	34,359,738,368	Medium Gamma	
38	+324	Q4		u_6	$1/128^6 = 2.27 \times 10^{-13}$	4.398 quadrillion	High-energy Gamma	
39	+342	Q4		u_7	$1/128^7 = 1.78 \times 10^{-15}$	562 quadrillion	Ultra-high / Cosmic Gamma	Deep nuclear/relativistic
40	+360	Return	Return to u_0		Return to 1	1	Unity Return (Cycle Complete)	Closure to unpartitioned whole

Part V

Quadrant Operational Breakdown

Q2: Gravity Ground States (-360° to -180°)**Phases:** 0–9 (10 phases)**Depth level:** u_1 dominant**Character:** Surface partition operations, gravitational ground states via dcos decompression. Shallow depths for long-wave propagation.**Electromagnetic bands:**

- Radio frequencies (ELF through L-band)
- Long wavelengths (km to cm range)
- Low partition depth (simple compression states)
- Minimal matter interaction (passes through easily)

Key marker: AM Radio at -270° (phase partition 5) — human communication baseline in 1D/2D surface states.**Q1: Quantum Baseline (-180° to 0°)****Phases:** 10–19 (10 phases)**Depth level:** u_2 dominant**Character:** Matter unit allocation via dsin overflow, EM ignition at deep 2D/3D transition.**Electromagnetic bands:**

- Microwave through Far-Infrared
- Medium wavelengths (cm to μm range)
- Molecular rotation and vibration coupling
- Thermal radiation emergence

Key markers:

- EM ignition at -168.75° : $\alpha^{-1} = 137$ calibration in 2D partition
- CMB peak at -126° (phase 13): Universal background in 3D volume states

Q3: Thermodynamic Flows (0° to $+180^\circ$)**Phases:** 20–29 (10 phases)**Depth level:** u_3 dominant**Character:** Volume partition operations, csin compression begins, entropy as deeper nesting.**Electromagnetic bands:**

- Infrared through Ultraviolet
- Short wavelengths (nm range)
- Atomic electron transitions
- Chemical bond breaking (UV)

Key markers:

- Seam at 0° (phase 20): $d\sin = c\sin =$ perfect balance in 3D
- Visible light ($+108^\circ$ to $+126^\circ$): Human perception window in 3D/4D hypervolume

Q4: Nuclear Compression ($+180^\circ$ to $+360^\circ$)

Phases: 30–40 (11 phases including return)

Depth level: u_4 and deeper (u_5+)

Character: Maximum density partitions, ccos compression to unity return. Deep dimensions for high-energy states.

Electromagnetic bands:

- X-rays through Gamma rays
- Very short wavelengths (pm to fm range)
- Penetrating radiation
- Nuclear processes and pair production

Key markers:

- EM absorption at $+191.25^\circ$: Thermal equilibrium in hypervolume
- Return to unity at $+360^\circ$ (phase 40): Cycle closure in infinite D limit

Part VI

Inverse Threading Mechanics

Pure Inverse Ratios Only — Walk-Through

All properties emerge from partitioning the whole — no addition:

- Start with unity: $\sqrt{1} = 1$
- Thread to first depth: $u_1 = 1/128$, closure $128 \times 1/128 = 1$
- Compound to $u_2 = 1/128^2$, closure $16384 \times 1/16384 = 1$
- Each photon at a phase is $1/n$ partition reconstituting to 1.

Unity conservation example: $(u_1/u_4)/(u_2/u_3) = 128^2 = 16384$ as ratio of depths.

Physical Constants as Geometric Invariants

Constant	Partition Derivation
$\alpha^{-1} = 137$	Gearbox difference at Q1 bin 16: 201 - 64
$T_{\text{CMB}} = 128/47$	Absorption overflow: 64 - 17 = 47
$E/T = 6016$	Cosmic threading link: partition ratio across depths
Entropy $S \approx 6.238$	$\ln 512$ — ways the whole threads the grid

Part VII

Comparison to Standard Physics and Operational Notes

Frequency Mapping Walk-Through

The 40-phase structure maps to observed frequencies via inverse octaves:

- Shallow u_1 : Low freq radio (long λ)
- Deeper u_n : Higher freq (short λ) as $f \propto 128^n$

Example alignment: AM at phase 5 ($\sqrt{5}D$), CMB at phase 13 ($\sqrt{13}D$).

Frequency Comparison Table

Band	Phase	Standard Frequency	Dimension
AM Radio	5 (-270°)	540–1600 kHz	$\sqrt{5}D$
CMB Peak	13 (-126°)	160.2 GHz	$\sqrt{13}D$
Visible (Red)	26 ($+108^\circ$)	430–480 THz	$\sqrt{26}D$
Visible (Violet)	27 ($+126^\circ$)	668–789 THz	$\sqrt{27}D$
Soft X-ray	30 ($+180^\circ$)	$\sim 3 \times 10^{16}$ Hz	$\sqrt{30}D$
Gamma rays	35–40	$> 10^{19}$ Hz	$\sqrt{35}D$ to $\sqrt{40}D$

Resolution Independence

Constants like α^{-1} are geometric — hold at any R, as ratios of partitions.

Operational Implementation — Binary Addressing Walk-Through

Each photon addressed by partition configuration (68-bit total):

- Q (2 bits): Quadrant partition code
- B (7 bits): Bin partition number (1–128)
- And so on — all pure binary divisions of the whole.

Future Extensions Walk-Through

Extend to deeper u_n (Q5+): partition further for Planck-scale physics, maintaining closure.

Inverse Spherical Dual Hemisphere Quantum Mechanics — Electromagnetic Full Spectrum Summary

Starting from $\sqrt{1} = 1D$ — partitions to $\sqrt{512} = 512D$

40 phases of 18° each = $720^\circ = 1$ sphere

Depth ladder $u_n = 1/128^n$ with closure to 1 at every level

Quadrants in dimensional progression: Q2 (1D/2D radio) → Q1 (2D/3D microwave) → Q3 (3D visible) → Q4 (4D+ gamma)

Bin-16 constants: $\alpha^{-1} = 137 (\sqrt{168.75D})$, $T_{\text{CMB}} = 128/47 (\sqrt{191.25D})$

Pure inverse partitioning — the whole divides itself,
reconstitutes exactly.

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